MIL-A-82430A(OS)
10 April 1968
Superseding
MIL-A-82430(OS)
10 May 1966

MILITARY SPECIFICATION

ACTUATOR ASSEMBLY

(For Torpedo Mk 46 Mod 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers requirements for the procurement of one type of actuator assembly (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-M-82433	Motor, Direct Current

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL 2540065	Actuator Assembly, and all documents listed thereon

FSC 1355

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DL 1330402	Actuator Assembly, Shipping (Packing) Condition and all documents listed thereon
DL 2526751	Afterbody Assembly and all documents listed thereon
2540214	Test Fixture
2540215	Torque Gage Assembly
2540260	Actuator Assembly Test Set
2540229	Actuator Assembly Test Stand
2540237	Wrench, Torque, T-Handle
2540238	Wrench, Torque

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using procedures proposed for the production lot.

(b) Consist of three actuator assemblies.

(c) Be tested as specified in Section 4 herein for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the actuator assembly shall be in accordance with requirements specified herein and with applicable documents listed in DL 2540065.

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3.3 Performance and product characteristics

NOTE: Numerical values tabulated in this specification are applicable to tests conducted in full accordance with 4.7. Minimum and maximum values apply to absolute values only (i.e., -15.6 volts direct current (vdc) is greater than -15.1 vdc).

3.3.1 Description

3.3.1.1 Function. The actuator assembly provides three axis (yaw, pitch, roll) hydrodynamic control of a torpedo through rotational actuation of the steering rudders and elevators. The actuator is an integral wired assembly consisting of three identical subassemblies. Each subassembly is a worm gear drive unit powered by a reversible permanent magnet dc motor with integral gearhead reducer in accordance with MIL-M-82433. The upper and lower drive units each operate a rudder (fin assembly) independently; the elevator drive unit operates two elevators (fin assemblies) which are interconnected by a clamping yoke.

3.3.1.2 Configuration. The actuator assembly shall conform to the requirements of the drawings and specifications listed on DL 2540065.

3.3.1.3 Functional operation. The external control circuitry provides the input signals to the actuator assembly. The elevator drive unit actuates the elevators for torpedo pitch control, the direction of rotation determined by the polarity of the pitch signal.

3.3.1.3.1 The upper and lower drive units actuate the upper and lower rudders respectively, the direction of rotation being determined by the polarity of the yaw signal. Roll control is obtained by reducing the speed of either the upper or lower drive unit, dependent on the polarity of the roll signal. The speed change is achieved by superimposing an interruption at 500 cps on the normal dc yaw signal. This speed change gives a difference in angular position of the rudders, thereby providing roll torque to the torpedo body.

3.3.1.3.2 The control fins move at constant rates in accordance with the input signal given until either a signal reversal occurs or the limit of travel is reached, at which time the signal power is disconnected by a limit switch until signal reversal does occur.

3.3.2 Performance requirements

3.3.2.1 Input signal. The input signal for normal fin rates shall be as shown in Table 1:

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Table I

Input Signal for Normal Fin Rates

Drive Unit	Direction of Output Shaft	Input Signal Connector Pin (J-5620)			Torpedo Reaction
		+36 ±1 VDC	Common	-36 ±1 VDC	
Upper Rudder	CW	C	B		Left Turn
	CCW		B	A	Right Turn
Elevator	CW		F	E	Pitch Up
	CCW	G	F		Pitch Down
Lower Rudder	CW	L	K		Right Turn
	CCW		K		Left Turn

3.3.2.2 Insulation resistance. Insulation resistance between all connector pins and ground (worm gear housings) shall be not less than 100 megohms at 200 vdc.

3.3.2.3 No load current. The current required to operate each drive unit at a nominal voltage of 36 ±1 vdc under zero load conditions shall be as follows:

Drive Unit	Max. Current at No Load
Elevator	235 milliamperes (ma)
Upper Rudder	875 ma
Lower Rudder	875 ma

3.3.2.4 Nominal output shaft rate, no load. The nominal output shaft rate (angular velocity) shall be 23 degrees per second under no load condition at 36 ±1 vdc input signal, for all drive units in either direction.

(a) Measurement of output shaft rate, no load. The output shaft rate measured using Test Fixture 2540214, as prescribed in 4.7.3.4, shall be within the range of 23 ±5 degrees per second for all drive units. The range is provided to include allowable variations in test equipment and calibration error.

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(b) Difference in output shaft rates for upper and lower drive units. During the conduct of each quality conformance test, the difference in output shaft rates between upper and lower drive units shall not exceed three degrees per second in the following combinations:

(1) Lower CCW rate minus upper CW rate

(2) Lower CW rate minus upper CCW rate

3.3.2.5 Acceleration time (motor time constant). The time required for each drive unit to accelerate from rest to 63 percent of steady state output velocity shall not exceed 15 milliseconds, under no load conditions and at 36 ±1 vdc input signal.

3.3.2.6 Output shaft travel. The output shaft travel from null or zero fin position to the point of limit switch operation in either direction shall be 10.0 ±0.5 degrees for all three drive units.

3.3.2.7 Backlash at output shaft. Backlash measured at the output shafts shall not exceed 15 minutes of total angular freedom, with 5 inch-pounds torque applied.

3.3.2.8 Performance at rated load conditions. The performance of each drive unit under loaded conditions shall be controlled by a verification of motor performance under the test loads required in the quality conformance tests of Motor Specification MIL-M-82433(OS). In addition, the existence of excessive frictional losses due to binding or maladjustment in each assembly shall be controlled by performance of the no load current test (see 4.7).

3.3.2.9 Instrumentation for measurement of fin position. Wire wound potentiometers may be included in the assembly of each drive unit if required by the procuring activity. The installation and adjustments, together with drawings of the required mounting brackets and gearing, are shown in DL 2540065. The potentiometer output voltages shall be free of electrical noise and shall have no discontinuity when supplied with 15 ±1 vdc input power and with drive units operated at the normal rate of 23 ±5 degrees per second. The potentiometers and circuits shall also meet the insulation resistance requirements of 3.3.2.2.

3.3.3 Life. Life requirements shall be as specified for the following unit conditions (see 6.3.1).

3.3.3.1 Service life. The service life of the actuator assembly shall be a minimum of 7.5 hours of cumulative operating time through repeated duty cycles as specified in 3.3.3.2.

3.3.3.2 Duty cycle. The duty cycle shall consist of 10 minutes operating time followed by a minimum of 30 minutes nonoperating time. The

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following conditions shall exist during the operating time for each drive unit: An external switching circuit shall reverse the actuator motor input signal to obtain an operating frequency of 5 ±1 cycles per second with the signal voltage at 36 ±1 vdc. The external switching circuit itself shall have a reversing time of less than 5 milliseconds. Load applied to the output shaft shall increase linearly with angular rotation in each direction from the null or zero position at a rate of 20 ±2 lb-in. per degree. The null position is defined as the center of total travel between the externally applied limits and shall coincide with the point where the applied load changes direction. During life testing, provision shall be made to adjust symmetry of the switching circuit if required to maintain travel in each direction from the null position to within a difference of 0.5 degree.

3.4 Environmental requirements. Unless otherwise specified, the actuator assembly shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed to the environmental conditions specified in this section.

3.4.1 Vibration

3.4.1.1 Vibration, ready and ready non-operating condition. The actuator assembly, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table II. The actuator assembly shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45 degrees F to +140 degrees F. Vibration is defined in 6.3.2.

3.4.1.2 Vibration, packaged. The actuator assembly shall be capable of meeting the performance requirements after being subjected, in the packaged condition (see 5.1.1), to the vibrations shown in Table II. The actuator assembly shall be vibrated for 15 minutes in each of the three principal orthogonal axes of the package.

Table II

Vibration Schedule

Condition (See 6.3.1)	Double Amplitude (in.) Minimum	Acceleration (g's) Minimum	Frequency Range (cps)
Packaged	--	1.0	5 to 60
Ready nonoperating	0.2	--	5 to 17
and ready	--	3.0	17 to 500

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3.4.2 Shock. The actuator assembly, packaged in accordance with DL 1330402 (see 5.1.1) shall be capable of meeting performance requirements subsequent to withstanding the shock of being dropped on a nonyielding surface from a height of 36 inches. Additionally, the actuator assembly, in the condition shown in Table III, shall withstand shocks specified in Table III.

Table III

Shock Schedule

Condition (See 6.3.1)	Peak Magnitude (g's)	Duration (milliseconds)	Acceleration Direction*	Number of Impacts
Ready nonoperating	60 ±10%	8.0 to 10.0	3 fwd or aft axis 3 left or right axis 3 up or down axis	9
Ready	45 ±10%	18.0 to 20.0	Long. fwd axis	3
	85 ±10%	0.5 to 1.5	3 up or down axis 3 right or left axis	6
	300 ±10%	0.5 to 1.5	Long. aft axis	3

*See Figure 1 for axes of actuator assembly

3.4.3 Acceleration. The actuator assembly shall be capable of withstanding accelerations specified in Table IV.

Table IV

Acceleration Schedule

Condition (See 6.3.1)	Magnitude (g's)	Acceleration Direction
Ready nonoperating	25 ±10%	Longitudinal Forward
and ready	35 ±10%	Longitudinal Aft

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3.4.4. Temperature. The actuator assembly shall be capable of withstanding temperatures for the following specified conditions.

3.4.4.1 Nonoperating and ready Nonoperating. The actuator assembly shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, environmental temperatures between -45 degrees F and +140 degrees F. Temperature is defined in 6.3.5.

3.4.4.2 Ready. The actuator assembly shall meet performance requirements and product characteristics of this specification after exposure to temperatures between zero and +140 degrees F for a period wherein duration of the upper limit does not exceed four hours at any one time.

3.4.4.3 Operating. The actuator assembly, stabilized for one hour at any temperature between 0 degrees F and +140 degrees F, shall be capable of meeting the performance requirements.

3.4.5 Humidity. The actuator assembly shall withstand exposure to 95 percent relative humidity at +110 degrees F for eight hours without degradation of finish or functional capabilities.

3.4.6 Liquid splash. The actuator assembly shall be capable of operation through one duty cycle (3.3.3.2) while exposed to splashing by either the liquid propellant, as described in 3.2.6 of Motor Specification MIL-M-82433, by sea water, or by both together.

3.5 Workmanship. Workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the actuator assembly shall be classified as follows:

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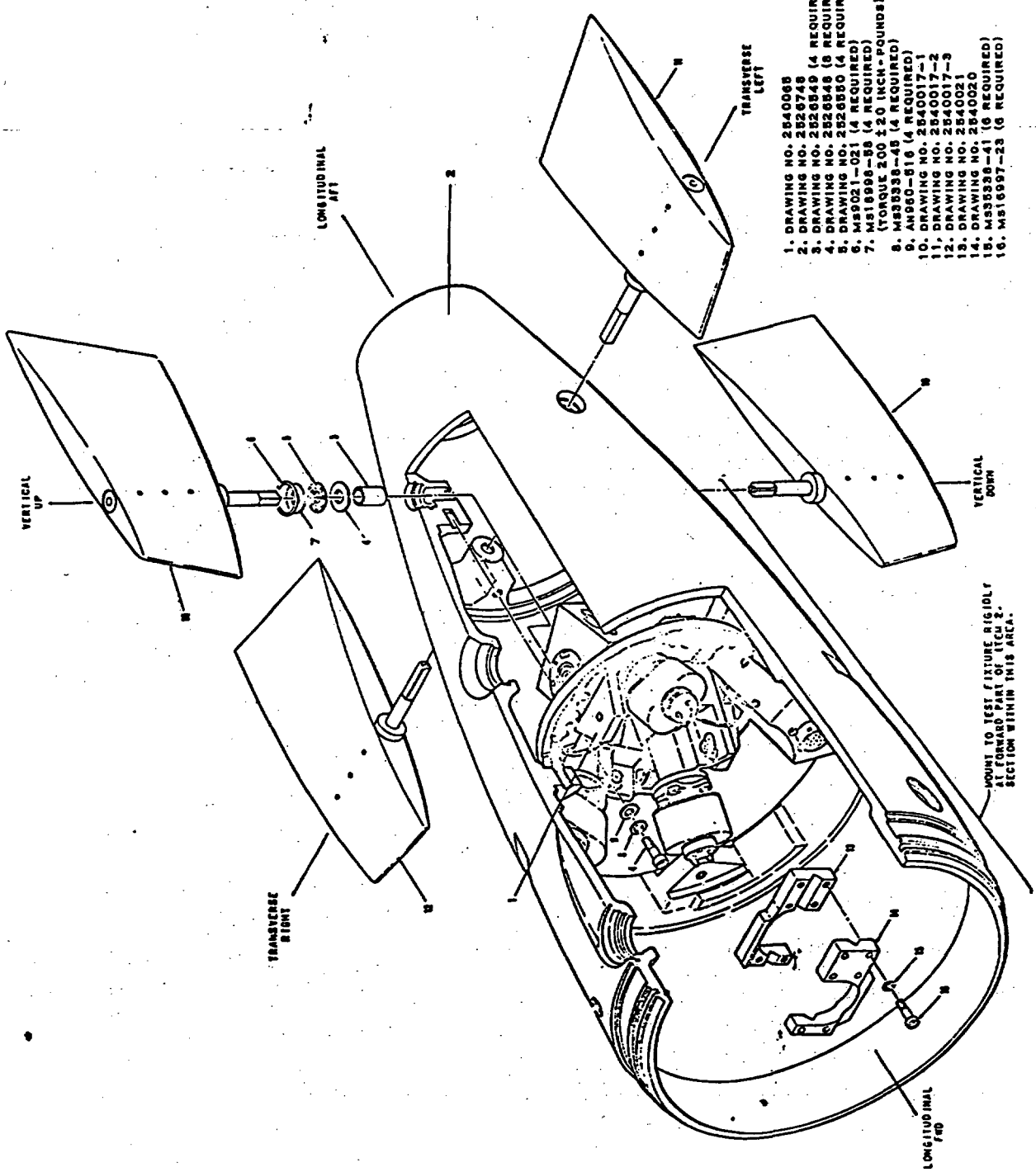


Figure 1. Axes of Actuator Assembly

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(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of three actuator assemblies, manufactured in accordance with 3.1, shall be subjected to preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the actuator assembly by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production randomly select one unit from each group of 30 units. Upon satisfactory compliance of samples selected from three consecutive groups shift to Step 2. Each sample so selected shall be numbered consecutively, starting with Sample Number 1.

(b) Step 2. Randomly select one unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively, starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit (see 6.3.8) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

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(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. Tests and examinations specified in 4.7 shall be conducted on each actuator assembly to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the actuator assembly shall be subjected to tests under following conditions:

(a) Temperature: Room ambient 65 degrees F to 95 degrees F

(b) Humidity (relative): Room ambient up to 95 percent

(c) Altitude: Normal ground atmospheric pressure

(d) Vibrations: None

4.5 Preproduction tests. The actuator assemblies within the preproduction sample (identified as actuator assemblies 1, 2, and 3) shall be subjected to tests specified in Table V. Certified inspection data covering results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests

(b) Run or test plan used

(c) Serial number of the actuator assembly

(d) Test results of preproduction tests (4.5.2)

(e) Report form, Quality Conformance Tests and Examinations for Actuator Assembly, for each test conducted after each environmental exposure as required in Table V.

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Table V

Preproduction Test Schedule

Test Sequence	Test Description	Actuator Assembly No.		
		1	2	3
1	Quality conformance tests and examinations (4.7)	x	x	x
2	Vibration (4.5.2.1).		x	x
3	Shock (4.5.2.2).	x		x
4	Acceleration (4.5.2.3).			x
5	Temperature, nonoperating (4.5.2.4.1).		x	x
6	Temperature, operating (4.5.2.4.2).	x		x
7	Humidity test (4.5.2.5).		x	x*
8	Life test (4.5.2.7).		x	x
9	Quality conformance tests and examinations (4.7).	x	x	x
10	Salt Water Spray (4.5.2.6)	x		x

*Quality conformance tests and examinations (4.7) to be performed after each of these tests.

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform preproduction tests specified herein:

4.5.1.1 A vibration table capable of vibrating the actuator assembly and necessary test fixtures at 3 g's maximum over a frequency range of 5 to 500 cycles per second with a maximum double amplitude of 0.2 inch and with provisions for controlling environmental temperature in the range of -45 degrees F to +140 degrees F.

4.5.1.2 A shock testing machine capable of applying shocks of up to 300 g's maximum at durations from 0.5 to 20 milliseconds.

4.5.1.3 A centrifuge capable of exerting acceleration forces up to 35 g's magnitude for period of one minute.

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4.5.1.4 A temperature chamber capable of programmed temperature variations over a range of -45 degrees F to +140 degrees F.

4.5.1.5 A humidity chamber capable of providing a constant humidity condition of 95 percent at 110 degrees F for extended periods up to eight hours.

4.5.1.6 A switching circuit and torsional load device capable of simulating the duty cycle specified in 3.3.3.2.

4.5.1.7 A salt water spray facility capable of producing conditions specified in 4.5.2.6.

4.5.1.8 A megohmmeter (200 vdc potential; accuracy ±5 percent).

4.5.1.9 Items of test equipment specified in Table VIII.

4.5.2 Detailed preproduction tests

4.5.2.1 Vibration

4.5.2.1.1 Nonoperating. The packaged actuator assembly (see 5.1.1) shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened in a suitable fashion to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 15 minutes at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.

4.5.2.1.2 Ready nonoperating and ready. The actuator assembly, including fin assemblies, bearing, and yokes, shall be installed in a torpedo afterbody or equivalent fixture. Installation shall be in accordance with afterbody assembly Drawing 2526751, observing tightening torque values specified for mounting of actuator assembly and installation of the yokes and fin assemblies. The afterbody or fixture, mounted on a support frame, shall be ridigly attached to the vibration equipment. This support frame shall be capable of transmitting the vibrations listed in Table II to the actuator assembly along the axes shown in Figure 1. Vibration will be conducted at three temperature conditions: the actuator assembly stabilized at room temperature, at -45 ±5 degrees F, and at +140 ±5 degrees F. The resonant modes for each of the three temperature conditions in each of the three axes shall be established by varying the frequency of applied vibration slowly from 5 to 500 cps, in accordance with Table II, for each temperature condition and in each axis. If more than four resonances are encountered for any one temperature condition on any one axis, the four most severe resonance points shall be chosen for the tests at resonance, for that one axis and that one temperature condition. The total vibration time in each axis shall be 180 minutes, with 60

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minutes of this time with the assemblage stabilized at -45 ±5 degrees F; 60 minutes with the assemblage stabilized at +140 ±5 degrees F; and 60 minutes at room temperature. The actuator assembly shall be vibrated for 10 minutes at each resonance point for each of the specified temperatures in each axis; the remainder of the 60 minutes shall be used in cycling per Table VI. The frequency rate during cycling shall be logarithmic such that 1/12 of the total vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range.

Table VI

Resonant and Cycling Schedule

No. of Resonances	Time at Resonance at Each Temperature in Each Axis, and at Each Resonance (Min)	Cycling Time at Each Temperature in Each Axis (Minutes)
0	0	60
1	10	50
2	10	40
3	10	30
4	10	20

4.5.2.2 Shock

4.5.2.2.1 Nonoperating. The packaged actuator assembly (see 5.1.1) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact surface which shall be essentially nonyielding stone, concrete, or similar material. The container, with the packaged unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3, and 6-1-4, and side 1 and face 5 as defined in Figure 2.

4.5.2.2.2 Ready nonoperating and ready. The actuator assembly, including fin assemblies, bearings, and yokes, shall be installed in a torpedo afterbody or equivalent fixture. Installation shall be in accordance with afterbody assembly Drawing 2526751, including tightening torque values specified for mounting of actuator assembly and installation of the yokes and fin assemblies. The afterbody or fixture shall be mounted so that a minimum of 90 percent of the applied shock will be transmitted to the actuator assembly. Apply the shocks listed in Table III for the conditions specified. Duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock.

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4.5.2.3 Acceleration. The actuator assembly, including fin assemblies, bearing, and yokes, shall be mounted in a torpedo afterbody or equivalent fixture and subjected to acceleration magnitudes and directions required by Table IV for one minute at room temperature for each condition specified.

4.5.2.4 Temperature

4.5.2.4.1 Temperature, nonoperating and ready nonoperating. The unpackaged actuator assembly shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

(a) Heat chamber to +140 +5 -0 degrees F in 2.5 ±0.5 hours.

(b) Maintain at +140 +5 -0 degrees F for four hours minimum, then within 2.5 ±0.5 hours cool chamber to -45 +0 -5 degrees F.

(c) Maintain at -45 +0 -5 degrees F for four hours minimum, then within 2.5 ±0.5 hours return chamber to room temperature.

Repeat this schedule for a minimum of three cycles.

4.5.2.4.2 Temperature, operating. The actuator assembly, mounted in a manner thermally representative of its final mounting in torpedo, shall meet test requirements as specified below:

(a) The actuator assembly shall be conditioned for at least one hour at a temperature of +140 degrees F ±5 degrees F prior to test. The assembly shall be operated through the duty cycle as specified in 3.3.2 in the +140 degree F environment. At conclusion of the duty cycle, allow 7 ±1/2 minute nonoperating period prior to performing insulation resistance test. Insulation resistance shall then be measured between all connector pins and ground (gear housing). Resistance shall be not less than 33 megohms at 200 vdc. The assembly shall then be allowed to cool and stabilize at room temperature. Insulation resistance shall then be in accordance with 3.3.2.2.

(b) The actuator assembly shall be conditioned for one hour at a temperature of 0 degrees ±5 degrees F prior to test. A test shall be conducted in the zero degree environment to prove initial performance capability. Applying the load specified in 3.3.3.2, each drive unit shall be energized with a signal input of 36 ±1 vdc, using the actuator assembly test set (see Table VIII). Each unit shall be operated starting from null or zero position to the limit of travel at 10 ±1/2 degrees in each direction. Total time from application of signal to completion of 10 degrees travel (point of limit switch operation) shall not exceed 0.8 second on the first attempt in each direction.

4.5.2.5 Humidity. The unpackaged actuator assembly shall be placed in a suitable test chamber and be subjected to a minimum of 95 percent relative humidity for a period of eight hours minimum at a temperature of +110 ±5 degrees F. After exposure, the actuator assembly shall be wiped

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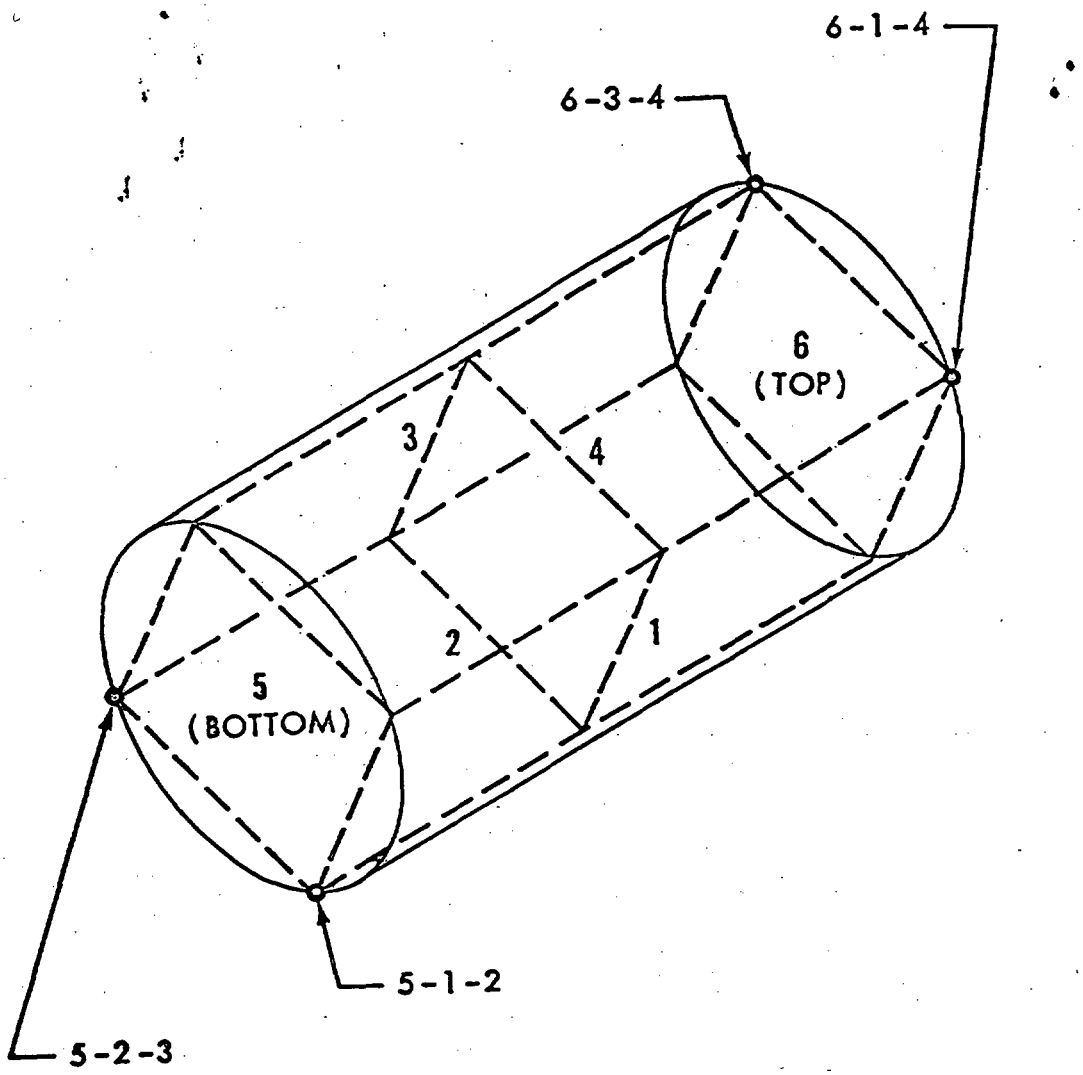


Figure 2. Definition of Faces and Corners of Shipping Container DL 1330402

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dry and placed in room environment up to 50 percent relative humidity for not more than one hour. Insulation resistance shall then be measured from each pin to ground (gear housing). Resistance shall exceed 10 megohms at 200 vdc. After standing in this room environment for not more than 24 hours, the insulation resistance shall be in accordance with 3.3.2.2.

4.5.2.6 Salt water spray test. The unpackaged actuator assembly shall be operated for one duty cycle as specified in 3.3.3.2 while exposed to a salt water spray test similar to the Rain Tests, Procedure II, as described in MIL-E-5272 (paragraph 4.10.2) with exceptions and/or special provisions as follows:

(a) The water introduced as rain shall be five percent sodium chloride solution prepared as in Federal Test Method Std. No. 151 Method 811 (paragraph 2.1.4) except that tap water may be used instead of distilled water. Alternatively, natural sea water may be used if available and clean.

(b) The rainfall rate shall be reduced to 2.0 ±0.5 inches per hour and the droplet diameters shall be mainly within the range of two to five millimeters.

(c) Variation of direction of rainfall from the vertical is not required but is permitted.

(d) The actuator assembly shall be placed upright in the chamber, with the aft-fwd axis horizontal.

(e) The assembly shall be exposed for a total period of 10 minutes.

4.5.2.6.1 Input power (or current) supplied to operate the three drive units shall be monitored throughout the 10 minute operating period while the assembly is exposed to the salt water spray. The input power shall not increase by more than 20 percent above the initial reading at the start of the operating period. Following the test the assembly should be sprayed thoroughly from various directions with warm fresh water to remove salt water from all areas, avoiding application of pressure to or immersion of motor shaft and drive coupling portions of the assembly. Although no further testing is required for units after completion of this test, the suggested washing procedure will help preserve the unit for possible future examination or disposition by the procuring activity.

4.5.2.7 Life test. The actuator assembly shall be subjected to a partial operating life test consisting of 45 duty cycles as specified in 3.3.3.2. The test shall be performed under such ambient conditions that the actuator assembly will cool to room temperature within a minimum of 30 minutes between each 10 minute operating period. Upon completion of the life test, the unit shall be subjected to the quality conformance tests and examinations (4.7), except that the limits for isolation current (4.7.3.3.1) shall be less than 4.25 microamperes and backlash (4.7.3.4.13) shall be less than 30 minutes of arc.

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4.6 Periodic production tests. Periodic production samples, selected in accordance with 4.3.2, shall be subjected to the tests of 4.5.

4.6.1 Periodic production test equipment. The items of test equipment shall be those specified for all tests of 4.5 and 4.7.

4.6.2 Periodic production test schedule. The periodic test sample shall be subjected to the tests noted in Table VII, in the order specified.

Table VII

Periodic Production Test Schedule

Test	Sample and Test Sequence		
	Sample No. 1,4, Etc.	Sample No. 2,5, Etc.	Sample No. 3,6, Etc.
Vibration (4.5.2.1)	1*		1*
Shock (4.5.2.2)		1*	3*
Acceleration (4.5.2.3)			2*
Temperature Nonoperating (4.5.2.4.1)	2		5
Hi-Temperature Operating (4.5.2.4.2,a)	3		4
Lo-Temperature Operating (4.5.2.4.2,b)		2	6
Humidity (4.5.2.5)		4	7
Life (4.5.2.7)			9
Quality conformance test (4.7)	4	3 and 5	8 and 10

*Indicates visual examination after test.

4.6.3 Periodic production acceptance criteria. Acceptance criteria shall be in accordance with 4.3.2.1 and 4.7.1.3.

4.7 Quality conformance tests and examinations. Unless otherwise specified each actuator assembly delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions

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4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in the order shown.

4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits is given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in the Reading column indicating that requirements were satisfactorily met.

4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units which have not been corrected will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.7.2 Quality conformance test equipment. The items of test equipment listed in Table VIII are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of the tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration. Record the verification of test equipment calibration.

4.7.3 Test procedure

4.7.3.1 Resistance test

4.7.3.1.1 Mount actuator assembly on actuator test stand. With T-handle Allen wrench (0.187 in.), tighten the three actuator mounting bolts (see Figure 3).

4.7.3.1.2 Connect interconnecting cable between actuator assembly and ACTUATOR INPUT located on actuator assembly test panel, hereafter referred to as test panel. Position controls on test panel as follows:

Control	Position
RESISTANCE	OFF
SHORT	UPPER

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Control	Position
MOTOR	UPPER
FUNCTION	RESISTANCE

Table VIII

Test Equipment

Item	Identification
Actuator Assembly Test Set	Dwg 2540260
Test Fixture	Dwg 2540214
Actuator Assembly Test Stand	Dwg 2540229
Torque Gage Assembly	Dwg 2540215
Oscilloscope: (dc to 15 mc pass band) 6 cm by 10 cm display graticule, (xl probe) 20 mv/cm deflection factor	Tektronix, Model 531 with 53/54 D plug-in preamplifier, or equivalent
Multimeter: (0 to 20 megohms resistance, 0 to 10 amps, 0 to 5000 volts ac and dc, ±3% full scale on dc, ±5% full scale on ac)	Simpson Model 260, or equivalent
Digital dc Voltmeter: (Range, 0.001 to 999 vdc; Accuracy ±0.2% of reading ±1 count) or DC Voltmeter 0-50 V ±0.25%	Hewlett-Packard Model 405CR or 405BR, or equivalent
T-Handle Torque Wrench (0.187 inches)	Dwg 2540237
Torque Wrench (0.125 inch)	Dwg 2540238
Rod to tighten locator screw 1/8 inch diameter maximum)	Commercial

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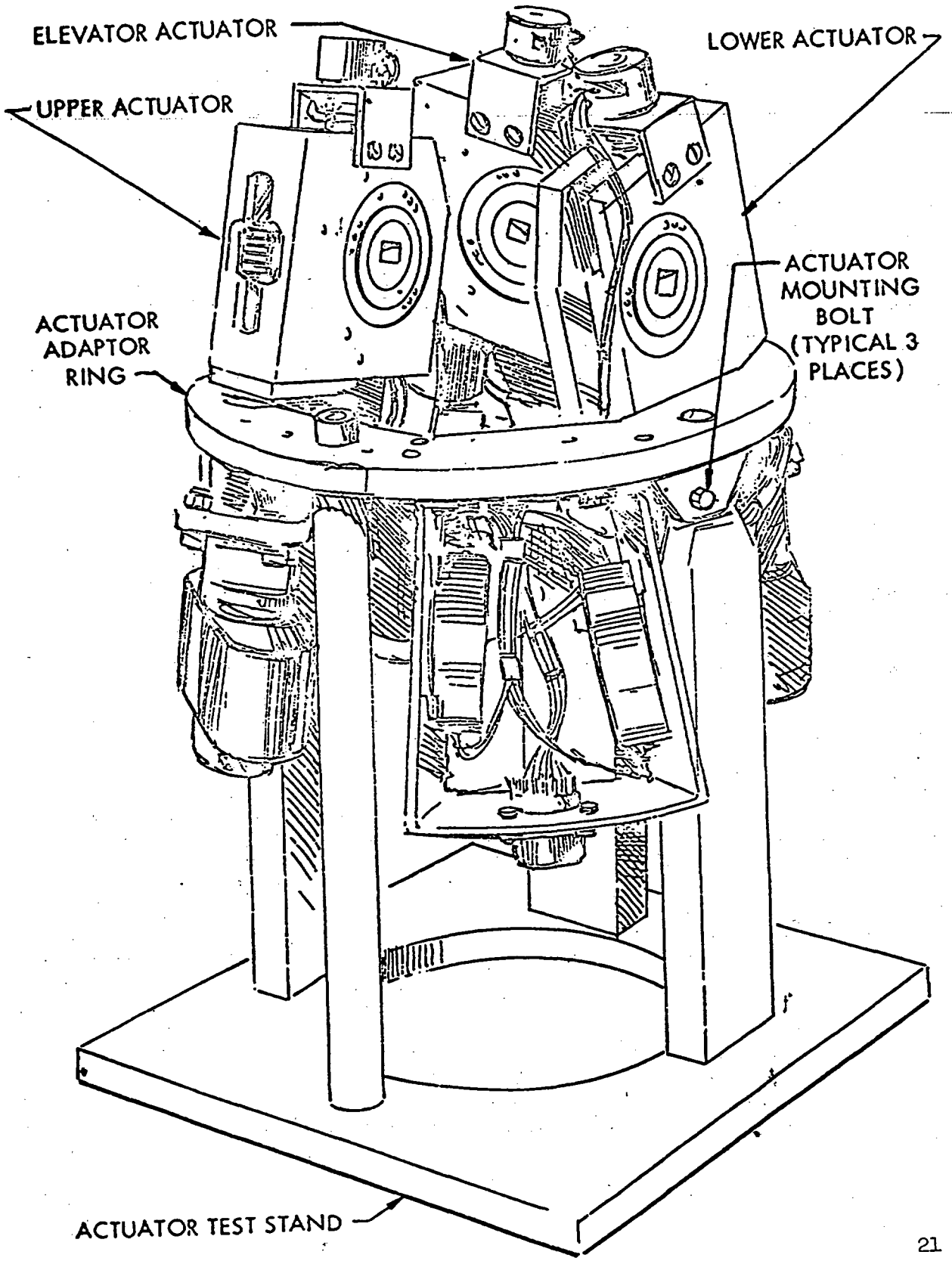


Figure 3. Actuator Assembly Mounted on Test Stand

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4.7.3.1.3 Connect multimeter to OHMS terminals located on test panel. Set multimeter to RX1 and "zero" needle indicator.

4.7.3.1.4 Position MOTOR switch to EL and record multimeter indication. The resistance shall be greater than 10 ohms.

4.7.3.1.5 Position MOTOR switch to LOWER and record multimeter indication. The resistance shall be greater than 10 ohms.

4.7.3.1.6 Position SHORT switch to LOWER and MOTOR switch to EL. Record multimeter indication. The resistance shall be greater than 10 ohms.

4.7.3.1.7 Position MOTOR switch to LOWER, SHORT switch to OFF, and RESISTANCE switch to MOTOR. Record multimeter indication. The resistance shall be 5 to 75 ohms.

NOTE: If motor circuit resistance exceeds 75 ohms on initial readings, it may be due to a temporary condition in which a surface film is formed on limit switch contacts due to environmental conditions. It is permissible to operate each drive motor from stop to stop, actuating each limit switch five times prior to reading motor circuit resistance as required in 4.7.3.1.7, 4.7.3.1.8, and 4.7.3.1.9.

4.7.3.1.8 Position MOTOR switch to EL, and record multimeter indication. The resistance shall be 5 to 75 ohms.

4.7.3.1.9 Position MOTOR switch to UPPER and record multimeter indication. The resistance shall be 5 to 75 ohms.

4.7.3.1.10 Position MOTOR and SHORT switches to LOWER. Set multimeter to RX100 scale and "zero" the meter.

NOTE: The tests specified in 4.7.3.1.11 through 4.7.3.1.15 shall be performed only on actuator assemblies which have instrumentation potentiometers installed.

4.7.3.1.11 Position SHORT switch to OFF, RESISTANCE switch to INST, and INSTRUMENTATION switch to LOWER. Record multimeter indication. The resistance shall be less than 2000 ohms.

4.7.3.1.12 Position INSTRUMENTATION switch to EL. Record multimeter indication. The resistance shall be less than 2000 ohms.

4.7.3.1.13 Position INSTRUMENTATION switch to UPPER and record multimeter indication. The resistance shall be less than 2000 ohms.

4.7.3.1.14 Position INSTRUMENTATION switch to INST and record multimeter indication. The resistance shall be 1500 to 1800 ohms.

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4.7.3.1.15 Position SHORT and MOTOR switches to LOWER, and RESISTANCE switch to OFF. Set multimeter to RX10,000 scale and "zero" meter. Position MOTOR switch to INST. Record multimeter indication. The resistance shall be greater than 5 megohms. Disconnect the multimeter from the test set.

4.7.3.2 Continuity test

4.7.3.2.1 Position controls on test panel as follows:

Control	Position
AC PWR	OFF
DC PWR	OFF
FUNCTION	ISOLATION
ISOLATION	OFF
INST	OFF
NORMAL	Depressed

4.7.3.2.2 Connect frame of actuator assembly to clip lead of interconnecting cable. Any screw head located on adapter ring of actuator assembly may be used for the connection. Set the LINE and OUTPUT switches on the power supply to the ON position.

4.7.3.2.3 Position AC PWR switch to ON. The red indicators shall illuminate. Record the indication.

4.7.3.2.3.1 Connect the digital voltmeter to the power supply output terminals and set the voltage to 42.5 ±0.1 vdc. Record that the voltage has been set. If actuator assembly includes instrumentation potentiometers, connect the digital voltmeter to the -15 VDC output terminals. The voltage shall be 15.0 ±0.2 vdc. Record the voltage.

4.7.3.2.4 Position and hold CONTINUITY switch in ON position. The indicator shall illuminate, showing continuity through the actuator circuit. Record the indication.

4.7.3.3 Isolation current test

4.7.3.3.1 Position DC PWR switch to ON. Position ISOLATION switch to INT. Record reading indicated on ISOLATION CURRENT meter. The reading shall indicate less than 0.4 microampere.

4.7.3.3.2 Disconnect clip lead of interconnecting cable from frame of actuator assembly.

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4.7.3.4 Limit switch setting, backlash, direction of output shaft rotation, shaft velocity, and instrumentation test. This test shall be performed on each actuator, in the order of UPPER RUDDER, ELEVATOR, and LOWER RUDDER actuators.

4.7.3.4.1 Position FUNCTION switch to ACTUATOR TEST.

4.7.3.4.2 Connect input A of oscilloscope to RATE OUTPUT. Position oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC SLOW
TRIGGER SLOPE	INT +
TRIGGER STABILITY	FULLY CLOCKWISE
TRIGGER LEVEL	FULLY CLOCKWISE
TIME/CM	100 MICROSECONDS
5X MAGNIFIER	OFF
MILLIVOLTS/CM	100
MV/CM MULTIPLIER	1
VARIABLE	CALIBRATED (FULLY CLOCKWISE)
INPUT MODE	A, DC

NOTE: Set trace intensity so that trace is thin but visible.

4.7.3.4.3 Loosen the three actuator mounting bolts to provide free play.

4.7.3.4.4 Mount test fixture carefully on UPPER actuator, as shown in Figure 4, as follows:

(a) Insert test fixture drive shaft into actuator output shaft and tighten test fixture locator screw by hand.

(b) With T-handle Allen wrench (0.187 in.), tighten actuator mounting bolt until lockwasher is compressed approximately one half.

(c) Tighten test fixture locator screw with drill rod wrench.

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(d) Tighten actuator mounting bolt with T-handle Allen wrench.

(e) With knurled-handle Allen wrench (0.125 in.), tighten expander screw in end of test fixture drive shaft.

4.7.3.4.5 Connect test fixture cable to TEST FIXTURE INPUT.

4.7.3.4.6 Depress SLOW locking switch.

4.7.3.4.7 Position ACTUATOR switch to UPPER.

4.7.3.4.8 Loosen clamp knob on dial indicator and position dial indicator so that test fixture large pointer may complete full travel without interference. Tighten clamp knob to hold dial indicator. (See Figure 4.)

4.7.3.4.9 Using the (SLOW) CW and CCW switches, position large pointer to line marked "0" on test fixture.

4.7.3.4.10 Loosen clamp knob and position dial indicator to contact indicator angle located on large pointer. Tighten clamp knob.

4.7.3.4.11 Position and hold torque gage assembly against force gage pin located on backlash pointer. Exert pressure on force gage against pin so that spring guide, located on force gage, aligns with scribed line marked "R". At same time set dial indicator to zero.

4.7.3.4.12 Exert pressure on torque gage assembly against pin so that spring guide aligns with scribed line marked "L". Read dial indicator to nearest minute of arc.

NOTE: Calibration of the dial indicator is directly in minutes of arc. Note total travel of dial indicator carefully, to ensure correct reading.

4.7.3.4.13 Record indicated backlash. The backlash shall be less than 15 minutes of arc.

4.7.3.4.14 Loosen clamp knob on dial indicator and position dial indicator so that large pointer may complete full travel without restriction. Tighten clamp knob.

*4.7.3.4.15 Depress (SLOW) CW switch until actuator motor runs to limit and stops. Position INST switch to INT. Record direction indicated by large pointer. For application of clockwise signal large pointer shall indicate CW. Record position of the large pointer. The large pointer shall position within the 9.5 to 10.5 degrees tolerance lines. Record the direction indicated by the instrumentation meter.

*Recordings of instrumentation meter (4.7.3.4.15 and 4.7.3.4.16) required only if instrumentation potentiometers are installed.

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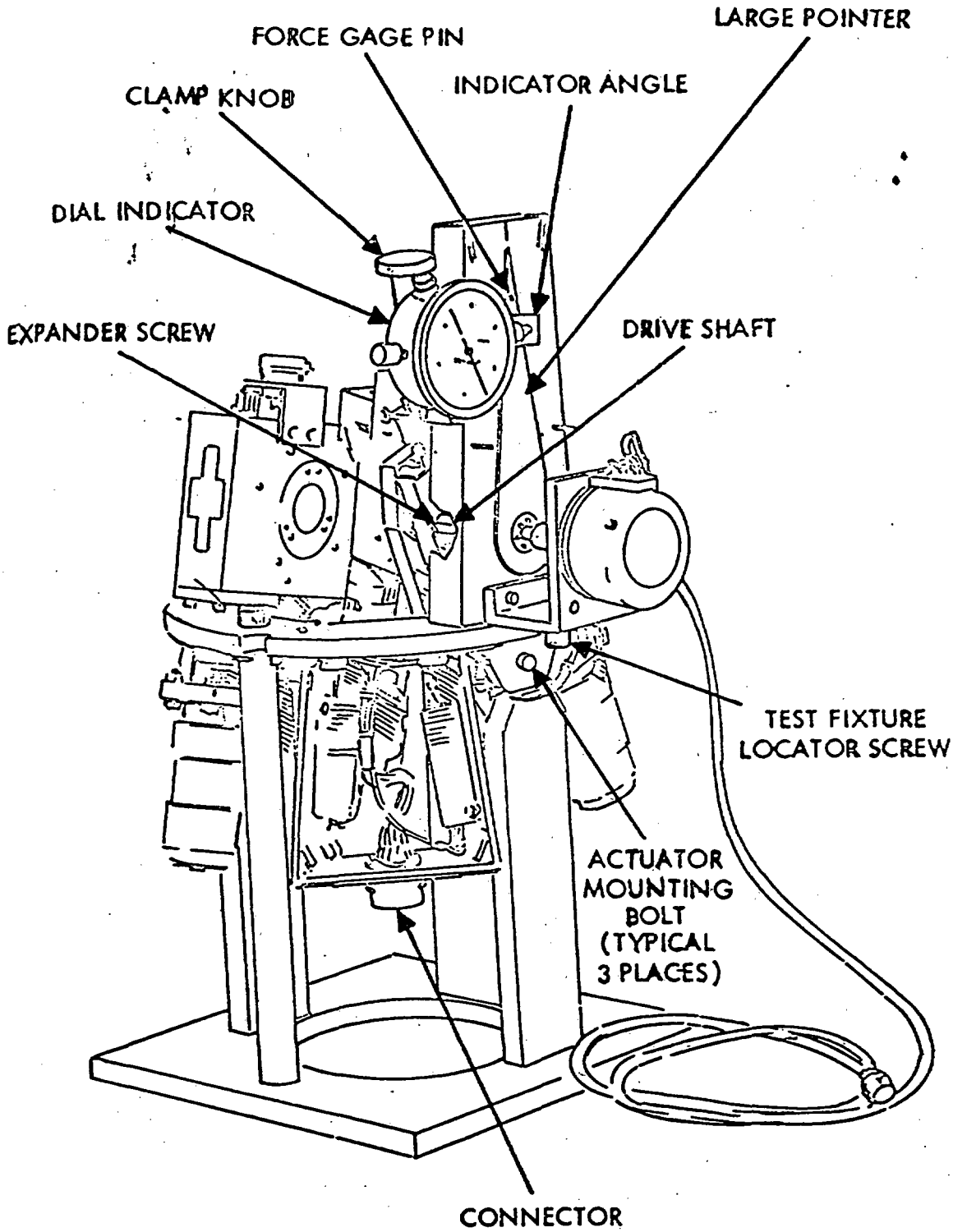


Figure 4. Test Fixture Mounting

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*4.7.3.4.16 Depress (SLOW) CCW switch until actuator motor runs to limit and stops. Record position of large pointer. The large pointer shall position within the 9.5 to 10.5 degrees tolerance lines. Position large pointer to "0". Record INSTRUMENTATION VOLTAGE OUTPUT meter indication. The indication shall be between -0.3 and +0.3 volts.

4.7.3.4 17 Depress NORMAL locking switch.

4.7.3.4.18 Using the vertical position control on the oscilloscope set trace on top line of graticule.

4.7.3.4.19 Depress (NORMAL) CCW switch until actuator runs to limit and stops. Depress (NORMAL) CW switch. Normal voltage will be applied to the actuator motor, which should run from limit to limit. Record maximum turn rate indicated on oscilloscope while actuator motor is moving over full range from limit to limit. The (NORMAL) CCW switch is used to return actuator motor to starting point. Calibration is 5 deg/sec/cm. The CW rate shall be between 18 and 28 deg/sec. The reading shall be taken with an accuracy of 0.5 deg/sec.

4.7.3.4.20 Using the vertical position control on the oscilloscope, set trace on top line of graticule.

4.7.3.4.21 Depress (NORMAL) CW switch until actuator motor runs to limit and stops.

4.7.3.4.22 Depress (NORMAL) CCW switch. Normal voltage will be supplied to the actuator motor which should run from limit to limit. Record maximum rate indicated on oscilloscope while actuator motor is moving over full range from limit to limit. The (NORMAL) CW switch is used to return the actuator motor to the starting point. Calibration is 5 deg/sec/cm. The CCW rate shall be between 18 and 28 deg/sec. The reading shall be taken with an accuracy of 0.5 deg/sec.

4.7.3.4.23 Carefully remove test fixture from UPPER actuator and install it on elevator actuator as shown in Figure 4 and as outlined in steps 4.7.3.4.4,(a) through (e).

4.7.3.4.24 Depress SLOW locking switch and place ACTUATOR switch in ELEV position.

4.7.3.4.25 Perform steps 4.7.3.4.8 through 4.7.3.4.23.

4.7.3.4.26 Carefully remove test fixture from elevator actuator and install it on LOWER actuator as shown in Figure 4 and as outlined in steps 4.7.3.4.4,(a) through (e).

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*Recordings of instrumentation meter (4.7.3.4.15 and 4.7.3.4.16) required only if instrumentation potentiometers are installed.

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4.7.3.4.27 Depress SLOW locking switch and position ACTUATOR switch to LOWER.

4.7.3.4.28 Perform steps 4.7.3.4.8 through 4.7.3.4.23.

4.7.3.4.29 Carefully remove test fixture from LOWER actuator motor

4.7.3.4.30 Remove test fixture cable from TEST FIXTURE INPUT and store test fixture.

4.7.3.4.31 Subtract the UPPER CW rate from the LOWER CCW rate. Subtract the UPPER CCW rate from the LOWER CW rate. In each case the difference shall be no greater than 3 deg/sec.

4.7.3.5 Steady state motor current test

4.7.3.5.1 Position ACTUATOR switch to UPPER. Depress CCW switch until actuator runs to limit and stops.

4.7.3.5.2 Depress (NORMAL) CW switch. Observe MOTOR CURRENT meter. Record the reading. Observe meter reading quickly since indicator positions only momentarily. Subtract 600 ma from meter reading. The steady state CW current shall be less than 235 ma. The reading shall be taken with an accuracy of 5 ma.

4.7.3.5.3 Depress (NORMAL) CCW switch. Observe MOTOR CURRENT meter. Record the reading. Observe meter reading quickly since indicator positions only momentarily. Subtract 600 ma from reading. The CCW steady state current shall be less than 235 ma.

4.7.3.5.4 Position ACTUATOR switch to EL.

4.7.3.5.5 Repeat steps 4.7.3.5.2 and 4.7.3.5.3. Do not subtract 600 ma from reading.

4.7.3.5.6 Position ACTUATOR switch to LOWER.

4.7.3.5.7 Repeat steps 4.7.3.5.2 and 4.7.3.5.3.

4.7.3.6 Motor time constant test

4.7.3.6.1 Connect input A of oscilloscope to MOTOR OUTPUT terminals. Position oscilloscope controls as follows in order shown:

Control	Position
INPUT SELECTOR	A, DC
TRIGGER MODE	AC SLOW
TRIGGER SCOPE	INT +

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Control	Position
TRIGGER STABILITY	FULLY CW
TRIGGER LEVEL	FULLY CW
TIME/CM	500 MICRO SEC
5X MAGNIFIER	OFF
MILLIVOLTS/CM	10
MV/CM MULTIPLIER	2
VARIABLE	CALIBRATED (FULLY CW)

A continuous sweep should appear on the oscilloscope. Turn sweep stability control on oscilloscope in a CCW direction until trace just disappears. Depress (NORMAL) CW switch to cause actuator motor to run to limit and stop. Continue depressing and releasing (NORMAL) CW switch and adjust triggering level control in a CCW direction until a single sweep is obtained when (NORMAL) CW switch is depressed.

4.7.3.6.2 Position ACTUATOR switch to UPPER. The following steps apply to the UPPER actuator motor:

4.7.3.6.3 Depress (NORMAL) CCW switch until actuator motor runs to limit and stops.

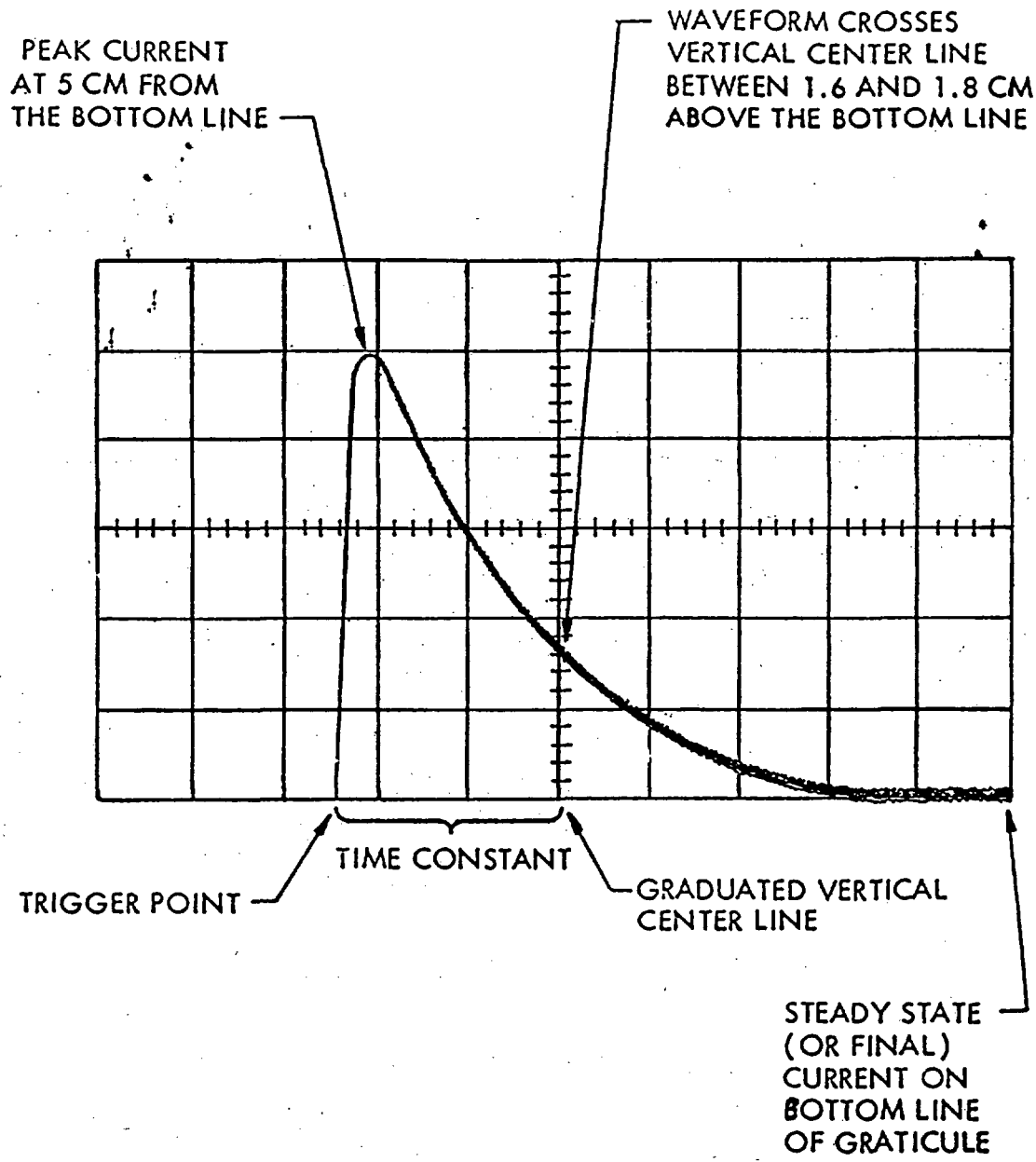
4.7.3.6.4 Position TIME/CM switch on oscilloscope to 50 MILLISEC.

4.7.3.6.5 Depress (NORMAL) CW switch. Using the VERTICAL POSITION, MV/CM MULTIPLIER, and VARIABLE controls on the oscilloscope, position steady state motor current trace at bottom line on graticule and peak motor current trace 5 cm above bottom line on graticule. (See Figure 5.) Use (NORMAL) CCW switch to return actuator motor to starting position.

4.7.3.6.6 Position TIME/CM switch on oscilloscope to 5 MILLISEC.

4.7.3.6.7 Position HORIZONTAL POSITION control on oscilloscope so that trace crosses graduated vertical center line between 1.6 and 1.8 cm above bottom line when (NORMAL) CW switch is depressed. (See Figure 5.) Record distance from triggering point to vertical center line. Calibration is 5 ms/cm. The time constant shall be less than 15 ms. The reading shall be taken with an accuracy of 1.0 ms.

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NOTE:
PEAK AND STEADY STATE CURRENTS ARE ADJUSTED BY USING THE VARIABLE MV/CM MULTIPLIER AND THE VERTICAL POSITION CONTROLS ON THE OSCILLOSCOPE. ADJUSTING THE WAVEFORM CROSSING POINT IS DONE LAST, USING THE HORIZONTAL POSITION CONTROL ON THE OSCILLOSCOPE.

Figure 5. Typical Waveform, Motor Time Constant

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4.7.3.6.8 Depress (NORMAL) CW switch until actuator motor runs to limit and stops.

4.7.3.6.9 Position TIME/CM switch on oscilloscope to 50 MILLISEC.

4.7.3.6.10 Depress (NORMAL) CCW switch. Using VERTICAL POSITION, MV/CM MULTIPLIER and VARIABLE controls on oscilloscope, position steady state motor current trace at bottom line on graticule and peak motor current trace 5 cm above bottom line on graticule. (See Figure 5.) Use (NORMAL) CW switch to return actuator motor to starting position.

4.7.3.6.11 Position TIME/CM switch on oscilloscope to 5 MILLISEC.

4.7.3.6.12 Position HORIZONTAL POSITION control on oscilloscope so that trace crosses graduated vertical center line between 1.6 and 1.8 cm above bottom line when (NORMAL) CCW switch is depressed. (See Figure 5.) Record distance from triggering point to vertical center line. (Calibration is 5 ms/cm.) The time constant shall be less than 15 ms. The reading shall be taken with an accuracy of 1.0 ms.

4.7.3.6.13 Position ACTUATOR switch to LOWER. The following steps apply to the LOWER actuator motor.

4.7.3.6.14 Repeat steps 4.7.3.6.3 through 4.7.3.6.12.

4.7.3.6.15 Position ACTUATOR switch to EL position. The following steps apply to the elevator actuator motor.

4.7.3.6.16 Repeat steps 4.7.3.6.3 through 4.7.3.6.12.

NOTE: The tests contained in 4.7.3.7 shall be performed only on actuator assemblies which have instrumentation potentiometers installed.

4.7.3.7 Instrumentation quality test

4.7.3.7.1 Connect input A of oscilloscope to ground post located on oscilloscope. Position oscilloscope controls as follows:

Control	Position
INPUT SELECTOR	A, DC
TRIGGER MODE	AC SLOW
TRIGGER SLOPE	INT +
TRIGGER STABILITY	FULLY CW
TRIGGER LEVEL	FULLY CW

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Control	Position
TIME/CM	100 MILLISECONDS
5X MAGNIFIER	OFF
MILLIVOLTS/CM	1000
MV/CM MULTIPLIER	5
VARIABLE	CALIBRATED (FULLY CW)

A continuous sweep should appear on oscilloscope. Position reference trace (0.0 volts) on bottom line of graticule. Connect input A of oscilloscope to INST OUTPUT terminals. Turn TRIGGER STABILITY control CCW until trace just disappears. Position INST switch to EXT. Depress (NORMAL) CW and CCW switches to run actuator motor to limit in both directions. Position TRIGGER LEVEL control to a position that presents a trace at same time actuator motor begins to run.

4.7.3.7.2 Position ACTUATOR switch to UPPER.

4.7.3.7.3 Depress (NORMAL) CW and CCW switches. Check instrumentation for intermittent or improper operation and noise. A normal trace should slope across the graticule from approximately 10 to 30 volts as the actuator motor runs to both limits. (Calibration is 5 volts/cm.) Improper operation is characterized by deviations from the slope, absence of the slope or by the trace going to 0.0 volts while the actuator motor is running or when it reaches limits. (See Figure 6.) Noise is characterized by spikes appearing on the trace while the actuator motor runs to both limits. There shall be no indication of improper or intermittent operation. There shall be no indication of noise on the trace.

4.7.3.7.4 Position ACTUATOR switch to EL.

4.7.3.7.5 Repeat step 4.7.3.7.3.

4.7.3.7.6 Position ACTUATOR switch to LOWER.

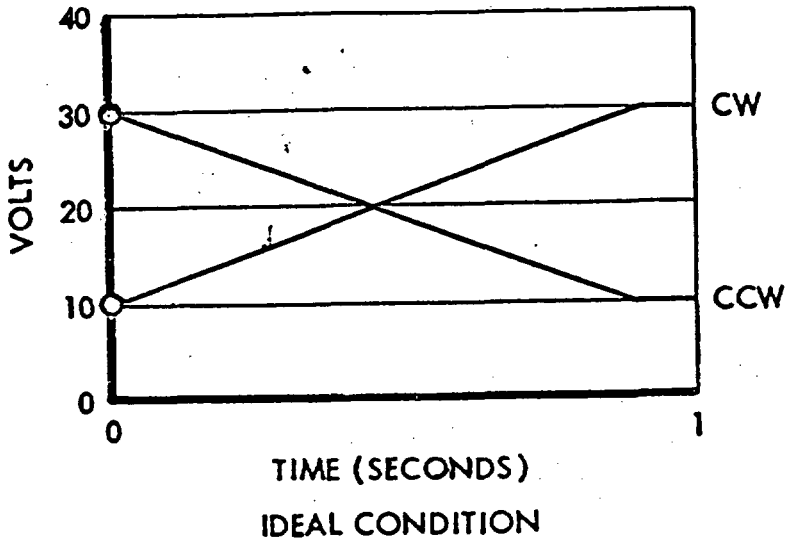
4.7.3.7.7 Repeat step 4.7.3.7.3.

4.7.3.7.8 Depress SLOW locking switch.

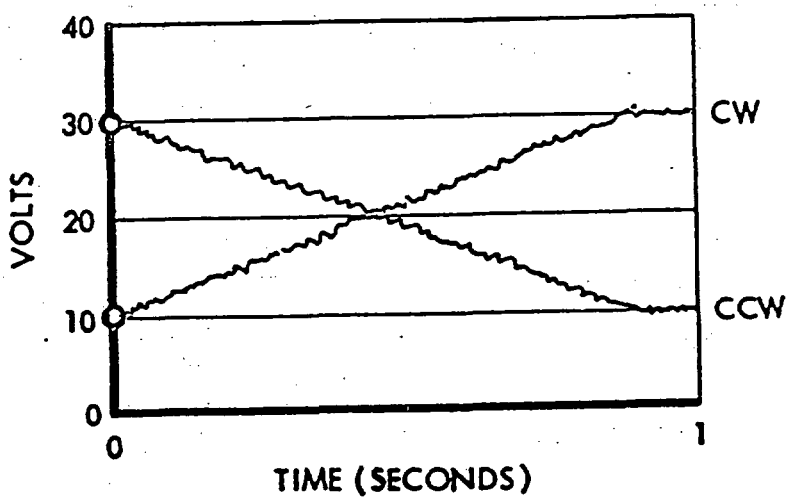
4.7.3.7.9 Position INST switch to INT.

*4.7.3.7.10 Position ACTUATOR switch to UPPER. Depress (SLOW) CW and CCW switches to position INSTRUMENTATION VOLTAGE OUTPUT indicator to read zero.

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IDEAL CONDITION



NOISE

THE CW SWITCH DISPLAYS A POSITIVE SLOPE FROM 10 TO 30 VOLTS.
THE CCW SWITCH DISPLAYS A NEGATIVE SLOPE FROM 30 TO 10 VOLTS.

Figure 6. Noise Display

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*4.7.3.7.11 Position ACTUATOR switch to EL. Depress (SLOW) CW and CCW switches to position INSTRUMENTATION VOLTAGE OUTPUT indicator to read zero.

*4.7.3.7.12 Position ACTUATOR switch to LOWER. Depress (SLOW) CW and CCW switches to position INSTRUMENTATION VOLTAGE OUTPUT indicator to read zero.

4.7.3.7.13 Depress NORMAL locking switch.

4.7.3.7.14 Position DC PWR switch to OFF.

4.7.3.7.15 Position AC PWR switch to OFF.

4.7.3.7.16 Remove input A of oscilloscope from INST OUTPUT terminals.

4.7.3.7.17 Disconnect interconnecting cable from Actuator Assembly and TEST FIXTURE INPUT.

4.7.4 Visual inspections

4.7.4.1 Identification marking

4.7.4.1.1 The nomenclature, control callout, and identification data shall be correct, clear, and legible. (Reference: Dwg DL 2540065.)

4.7.4.2 Interchangeability

4.7.4.2.1 The connector receptacle shall be correct and capable of accepting the mating unit without resort to selection, modification, or fitting. (Reference: Dwg DL 2540065.)

4.7.5 Process verification

4.7.5.1 Criteria stipulated in this paragraph and its subparagraphs are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.7.5.2 Lacing

4.7.5.2.1 Lacing and cable configuration shall be in accordance with procedures approved by the procuring activity. Specific method of verifying adequacy of cable lacing shall include visual and dimensional verification.

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*Center Actuators without instrumentation potentiometers by positioning large pointer to "0" (see 4.7.3.4.16).

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4.7.5.3 Soldering

4.7.5.3.1 Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.7.5.4 Cleanliness

4.7.5.4.1 The presence of foreign material in the assembly shall classify the unit as defective. Foreign material includes anything not intended to be present.

4.7.6 Verification of motor conformance. Verification of motor conformance to Motor Specification MIL-M-82433 shall be made by examination of Motor Quality Conformance Test Data which shall be submitted in reproducible copy with each actuator assembly to be delivered.

4.7.7 Verification of actuator subassembly worm gear backlash. Worm gear backlash shall be verified for each actuator subassembly serial number in accordance with Note 6 of actuator subassembly Drawing 2540064. The maximum and minimum backlash values obtained during inspection at subassembly level shall be recorded in paragraph 30.7 of the Quality Assurance Provisions.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2(b) and 6.4).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for Level A shipment shall be in accordance with the instructions contained in the applicable documents listed in DL 1330402.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2(c) and 6.4).

5.2.1 Level A. Multiples of the actuator assembly, preserved and packaged in accordance with 5.1.1, shall be packed in accordance with MIL-P-116, Method III.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The actuator assembly, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

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5.3 Marking

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with DL 1330402.

6. NOTES

6.1 Intended use and description

6.1.1 Intended use. The actuator assembly is intended for use in a torpedo.

6.1.2 Description. See 3.3.1.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Required data.

(e) Assigned activity for preproduction and periodic production tests.

(f) Any exceptions to this specification.

(g) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition

6.3.1.1 Nonoperating. The condition of the uninstalled actuator assembly when stored, transported, or handled (see 5.1.1).

6.3.1.2 Ready, Nonoperating. The condition of the actuator assembly when ready for assembly or assembled in a torpedo being issued, transported, or handled.

6.3.1.3 Ready. The condition of the actuator assembly when fully assembled with the end item and installed on the launching mechanism prior to operating.

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6.3.1.4 Operating. The condition of the actuator assembly when performing its required function.

6.3.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.3 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's) of duration in units of time, milliseconds (ms), and in the form of a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature. The temperature of the environment to which the actuator assembly may be exposed, expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. The pressure of the environment to which the actuator assembly may be exposed, expressed in pounds per square inch absolute (psia).

6.3.7 Humidity. The relative humidity of the environmnet to which the actuator assembly is exposed, expressed as a percent.

6.3.8 Defective unit. A unit shall be considered defective if it does not meet all examinations or tests specified, or deviations are found through visual examination after environmental exposure.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

(a) Level A: This level shall be used for those items which are to be shipped to intermediate destinations or stored under indeterminate conditions for redistribution anywhere.

(b) Level B: Not applicable.

(c) Level C: This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

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6.5 Storage life. The actuator assembly, preserved and packaged in accordance with Section 5, is intended to withstand an unattended storage life of five years.

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

ACTUATOR ASSEMBLY

DL 2540065

(For Torpedo Mk 46 Mod 1)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION	GOVERNMENT ACCEPTANCE
Contractor	Date of Acceptance
Contract	Accepted by
Equipment Serial No.	Title
Insp. Req. Satisfied	Agency

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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

ACTUATOR ASSEMBLY

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, and when the authorized certification and acceptance signatures are affixed, this report form or copies thereof shall accompany the actuator assembly and include the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the actuator assembly. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and shall be performed in the order shown.

20.2 Recording of results: Where space is provided to indicate the value of specific reading obtained, the specific reading will be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded.

Column I limits shall apply to the following conditions:

(1) Factory acceptance test

(2) Quality conformance test following adjustment or overhaul by an authorized F.I.R. repair activity.

Column II limits shall apply under the following conditions only:

(1) Backlash and Insulation Resistance requirements following a life test as specified in para 4.5.2.7

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(2) Actuator travel requirements for all quality conformance tests conducted after factory acceptance or acceptance by an authorized F.I.R. repair activity.

Where no limits are given in Column II the limits of Column I shall apply for all test conditions. If limits are not stated in either Limit column, the individual performing the test or inspection shall initial in the space provided in the reading column indicating that the requirements have been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

		REFERENCE	LIMITS COL I	COL II	READING
30.1	Test equipment	4.7.2	In Calibration		_____
30.2	Test procedure	4.7.3			
30.2.1	Resistance test motor circuits	4.7.3.1			
	Upper to elevator	4.7.3.1.4	Greater than 10 ohms		_____
	Upper to lower	4.7.3.1.5	Greater than 10 ohms		_____
	Lower to elevator	4.7.3.1.6	Greater than 10 ohms		_____
	Lower	4.7.3.1.7	5 to 75 ohms		_____
	Elevator	4.7.3.1.8	5 to 75 ohms		_____
	Upper	4.7.3.1.9	5 to 75 ohms		_____
	*Instrumentation circuits, lower	4.7.3.1.11	Less than 2000 ohms		_____
	*Elevator	4.7.3.1.12	Less than 2000 ohms		_____
	*Upper	4.7.3.1.13	Less than 2000 ohms		_____

*Data required only if instrumentation potentiometers are installed.

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			Limits		
30.2.1	(Continued)	Reference	Col. I	Col. II	Reading
	*Instrumentation circuit	4.7.3.1.14	1.5K to 1.8K		_____
	*Instrumentation circuit/motor circuit	4.7.3.1.15	Greater than 5 megohms		_____
30.2.2	Continuity test	4.7.3.2			_____
	Red indicator	4.7.3.2.3	Illuminates		_____
	Set 42.5V supply	4.7.3.2.3.1	42.5 ±0.1		_____
	15 vdc supply set		15.0 ±0.2		_____
	Continuity indicator	4.7.3.2.4	Illuminates		_____
30.2.3	Isolation current test	4.7.3.3			
	Isolation current	4.7.3.3.1	Less than 0.4 μa	Less than 4.24 μa (After Life)	_____
30.2.4	Rudder and elevator functional tests	4.7.3.4			
	UPPER RUDDER				
	Backlash	4.7.3.4.13	Less than 15' of arc	Less than 30' of arc (After Life)	_____
	Pointer indication	4.7.3.4.15	Must be CW		_____
	Pointer position CW		9.5° to 10.5°	9.0° to 11.0°	_____

*Data required only if instrumentation potentiometers are installed.

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30.2.4 (Continued)	Reference	Limits Col. I	Limits Col. II	Reading
Rudder and elevator functional tests				
UPPER RUDDER (Cont'd)				
*Instrumentation Meter		must be CW		
Pointer Position CCW	4.7.3.4.16	9.5° to 10.5°	9.0° to 11.0°	_____
*Instrumentation voltage		-0.3 to +0.3 vdc		_____
CW rate	4.7.3.4.19	18 to 28 deg/sec		_____
CCW rate	4.7.3.4.22	18 to 28 deg/sec		_____
ELEVATOR	4.7.3.4.26			
Backlash	4.7.3.4.13	Less than 15' of arc	Less than 30' of arc (After Life)	_____
Pointer indication	4.7.3.4.15	must be CW		_____
Pointer position CW		9.5° to 10.5°	9.0° to 11.0°	_____
*Instrumentation meter		must be CW		_____
Pointer position CCW	4.7.3.4.16	9.5° to 10.5°	9.0° to 11.0°	_____

*Data required only if instrumentation potentiometers are installed.

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30.2.4 (Continued)	Reference	Limits Col. I	Limits Col. II	Reading
*Instrumentation voltage		-0.3 to +0.3 vdc		
CW rate	4.7.3.4.19	18 to 28 deg/sec		
CCW rate	4.7.3.4.22	18 to 28 deg/sec		
LOWER RUDDER	4.7.3.4.29			
Backlash	4.7.3.4.13	Less than 15' of arc	Less than 30' of arc (After Life)	
Pointer indication	4.7.3.4.15	must be CW		
Pointer position CW		9.5° to 10.5°	9.0° to 11.0°	
*Instrumentation meter		must be CW		
Pointer position CCW	4.7.3.4.16	9.5° to 10.5°	9.0° to 11.0°	
*Instrumentation voltage		-0.3 to +0.3 vdc		
CW rate	4.7.3.4.19	18 to 28 deg/sec		
CCW rate	4.7.3.4.22	18 to 28 deg/sec		
Lower CCW rate minus upper CW rate		3 deg/sec max		

*Data required only if instrumentation potentiometers are installed.

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			Limits		
30.2.4	(Continued)	Reference	Col. I	Col. II	Reading
	Lower CW rate minus upper CCW rate		3 deg/sec max		_____
30.2.5	Steady state motor current	4.7.3.5			
	UPPER RUDDER				
	CW	4.7.3.5.2	Less than 235 ma		_____
	CCW	4.7.3.5.3	Less than 235 ma		_____
	ELEVATOR	4.7.3.5.5			
	CW	4.7.3.5.2	Less than 235 ma		_____
	CCW	4.7.3.5.3	Less than 235 ma		_____
	LOWER RUDDER	4.7.3.5.7			
	CW	4.7.3.5.2	Less than 235 ma		_____
	CCW	4.7.3.5.3	Less than 235 ma		_____
30.2.6	Motor time constant	4.7.3.6			
	UPPER RUDDER				
	Normal switch, CW	4.7.3.6.7	Less than 15 ms		_____
	Normal switch, CCW	4.7.3.6.12	Less than 15 ms		_____
	LOWER RUDDER	4.7.3.6.14			
	Normal switch, CW	4.7.3.6.7	Less than 15 ms		_____
	Normal switch, CCW	4.7.3.6.12	Less than 15 ms.		_____

*Data required only if instrumentation potentiometers are installed.

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		Reference	LIMITS Col. I	Col. II	Reading
30.2.6	(Continued)				
	ELEVATOR	4.7.3.6.16			
	Normal switch CW	4.7.3.6.7	Less than 15 ms		_____
	Normal switch, CCW	4.7.3.6.12	Less than 15 ms		_____
30.2.7	*Instrumentation quality	4.7.3.7			
	Instrumentation output actuator switch				
	UPPER	4.7.3.7.3	10 to 30 volts		_____
	EL	4.7.3.7.5	10 to 30 volts		_____
	LOWER	4.7.3.7.7	10 to 30 volts		_____
30.3	Visual inspections	4.7.4			
30.3.1	Identification marking	4.7.4.1.1			_____
30.3.2	Interchangeability	4.7.4.2.1			_____
30.4	Process verification	4.7.5			
30.4.1	Lacing	4.7.5.2.1			_____
30.4.2	Soldering	4.7.5.3.1			_____
30.4.3	Cleanliness	4.7.5.4.1			_____
30.5	Preparation for delivery	5.0			
30.5.1	Packaging	5.1.1			_____

*Data required only if instrumentation potentiometers are installed.

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	Reference	Limits Col. I	Limits Col. II	Reading
30.5.2 Packing	5.2.1			_____
30.5.3 Marking	5.3.1			_____
30.6 Verification of motor conformance	4.7.6			
30.6.1 Upper rudder DC motor				
S/N_____		MIL-M-82433 (OS)		_____
30.6.2 Lower rudder DC Motor				
S/N_____		MIL-M-82433 (OS)		_____
30.6.3 Elevator DC motor				
S/N_____		MIL-M-82433 (OS)		_____
30.6.4 Motor Quality Conformance Test Data attached for each of the three motors.				_____

	Reference	Limits	Reading
30.7 Verification of actr subassy backlash	4.7.7		
30.7.1 Upper rudder actr subassy			
S/N_____			
Backlash max		0°15'	_____
Backlash min		---	_____
Variation (max to min)		Less than 7'	_____

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	Reference	Limits	Reading
30.7.2	Lower rudder actr subassy.		
	S/N_____		
	Backlash max	0°15'	_____
	Backlash min	---	_____
	Variation (max to min)	Less than 7'	_____
30.7.3	Elevator actr subassy		
	S/N_____		
	Backlash max	0°15'	_____
	Backlash min	---	
	Variation (max to min)	Less than 7'	

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-82430A(OS) ACTUATOR ASSEMBLY (FOR TORPEDO MK 46 MOD 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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